

DAY 15: POPULATION VS SAMPLE

Data Analysis Mathematics Series — Sampling Techniques & Outlier Impact

SECTION 1: CORE THEORY / MAIN CONCEPT

1. Population vs Sample

Population (N): The complete set of all possible observations, individuals, or items that share a common characteristic and are the subject of a statistical study.

Population (N): Kisi bhi study ya research me jitne bhi possible elements, log ya items hote hain, un sabke complete collection ko Population kehte hain.

Sample (n): A smaller, manageable subset of data collected or selected from the population. It must reflect the characteristics of the entire group.

Sample (n): Poori population me se chuna gaya ek chota sa hissa (subset) jo poori population ko achhe se represent kare, use Sample kehte hain. Kyunki real world me sabka data nikalna impossible aur expensive hota hai.

2. Sampling Techniques

To ensure a sample is representative and unbiased, we use specific methods:

- **Simple Random Sampling:** Every member has an equal chance of selection (like a lottery).
- **Stratified Sampling:** The population is divided into subgroups (strata) based on traits (e.g., Age, Income), and samples are drawn randomly from each subgroup.
- **Systematic Sampling:** Selecting every k^{th} element from a structured list (e.g., every 10th customer).

Sample bina kisi bhedbhav (bias) ke sahi chuna jaye, uske liye 3 main tarike hote hain:

- **Simple Random Sampling:** Yeh bilkul lottery system jaisa hota hai. Population me se kisi ko bhi randomly chun liya jata hai, sabka barabar chance hota hai.
- **Stratified Sampling:** Isme population ko pehle unke traits (jaise Age, Gender, ya Income) ke basis par alag-alag groups (Strata) me baanta jata hai, fir har group se randomly sample liya jata hai.
- **Systematic Sampling:** Ek fixed interval par data select karna. Jaise list me se har 10th person ko sample me chunna.

SECTION 2: SPECIAL CONDITIONS & OUTLIER IMPACT

CONDITION 1: Safe & Ideal Selection

Scenario: The sampling technique is strictly random and representative. No outliers are mistakenly overrepresented.

Result: The Sample Mean ($\bar{x}$) mirrors the Population Mean (μ) very closely. The data analysis leads to correct business decisions.

Condition: Jab sampling ekdum perfect aur random ho, aur koi bhi outlier galti se sample me heavy weight na le paye.

Result: Is case me Sample Mean ($\bar{x}$) aur Population Mean (μ) ek dusre ke bohot kareeb hote hain, jisse aapka data analysis ekdum accurate result deta hai.

CONDITION 2: Flawed Selection with Outlier Bias

Scenario: Due to a small sample size or bad selection tracking, extreme values (Outliers) are selected.

Result: The sample becomes heavily skewed or distorted. The Sample Mean deviates drastically from the reality of the population, leading to highly misleading analysis.

Condition: Jab sample size bohot chota ho ya kismat kharab ho, aur galti se data ki extreme values (Outliers) sample me select ho jayein.

Result: Isse aapka sample completely skew (distort) ho jata hai. Sample Mean actual reality se bohot door chala jata hai aur aapka decision galat ho jayega.

SECTION 3: MATHEMATICAL FORMULAE & NUMERICAL COMPONENT

Mathematical Formulae:

$$\text{Population Mean } (\mu) = \sum X_i / N$$

$$\text{Sample Mean } (\bar{x}) = \sum x_i / n$$

Practical Case Study / Numerical Sawal

A jewelry manufacturing unit has 5 employees. Their monthly wages (in INR) are:

30,000 | 35,000 | 40,000 | 45,000 | 5,00,000 (Owner/Outlier)

1. Calculate the absolute Population Mean (μ).
2. **Flawed Sample Check:** If a sample of size $n = 2$ accidentally picks the worker with 30,000 and the Owner (5,00,000), calculate the Sample Mean ($\bar{x}$) and analyze the error.

Ek jewelry manufacturing unit me 5 log kaam karte hain. Unki monthly salary (INR me) ye hai:

30,000 | 35,000 | 40,000 | 45,000 | 5,00,000 (Owner ki salary, jo ki Outlier hai)

1. *Pehle poori Population ka Mean (μ) nikalye.*
2. **Galti ka test:** *Agar hum $n = 2$ ka chota sample lete hain aur galti se usme 30,000 wala employee aur 5,00,000 wala Owner aa jata hai, to Sample Mean ($\bar{x}$) nikalye aur dekhiye kya galti hoti hai.*

Step-by-Step Solution

Step 1: Population Mean (μ) Calculation

Total Elements (N) = 5.

$$\mu = (30000 + 35000 + 40000 + 45000 + 500000) / 5 = 650000 / 5 = 1,30,000$$

Meaning: Mathematically company ka average salary 1.3 Lakh aaya, jo ki us ek akele outlier (owner) ki wajah se pehle hi bohot bada dikh raha hai.

Step 1: Population Mean (μ) Calculation

Kul elements (N) = 5 hain.

$$\mu = (30000 + 35000 + 40000 + 45000 + 500000) / 5 = 650000 / 5 = 1,30,000$$

Matlab: Mathematically company ka average salary 1.3 Lakh aaya, jo ki us ek akele outlier (owner) ki wajah se pehle hi bohot bada dikh raha hai.

Step 2: Flawed Sample Mean ($\bar{x}$) under Condition 2

Sample Elements (n) = 2. Values selected: 30,000 and 5,00,000.

$$\bar{x} = (30000 + 500000) / 2 = 530000 / 2 = 2,65,000$$

Analysis: Sample mean shows 2.65 Lakhs. If an analyst uses this sample blindly, they will wrongly conclude that every worker in this factory is extremely rich. This is a severe Sampling Error.

Step 2: Flawed Sample Mean ($\bar{x}$) under Condition 2

Sample size (n) = 2 hai. Chune gaye log: 30,000 aur 5,00,000.

$$\bar{x} = (30000 + 500000) / 2 = 530000 / 2 = 2,65,000$$

Analysis: Sample mean aaya 2.65 Lakh! Agar aap is sample ko blindly sach maan loge, to aap bologe ki "Is factory ka har worker bohot ameer hai", jo ki ekdum bakwas aur galat data analysis hoga. Is galti ko hi **Sampling Error** kehte hain.

SECTION 4: DATA ANALYST RESOLUTIONS (SOLUTION)

To handle Condition 2 and avoid outlier errors, professional analysts apply:

- **Measure Shift:** Use **Median** instead of Mean when outliers exist (The median of the population above is 40,000, which is highly accurate).
- **Law of Large Numbers:** Increase sample size (n). A larger base dilutes the heavy mathematical impact of a single outlier.

Is tarah ki galti (Condition 2) se bachne ke liye professional Data Analysts ye do kaam karte hain:

- **Median Ka Use:** Jab data me aise bade outliers hon, to Mean ki jagah **Median** (middle value) ka use karo. Upar diye gaye data ka median 40,000 aayega, jo ki ekdum accurate representation hai workers ki salary ka.
- **Sample Size Bada Karo:** Hamesha sample size (n) ko bada rakho. Agar sample me 50 ya 100 log honge, to usme ek-adh outlier aa bhi gaya to uska mathematical asar baki bada data dilute (khatam) kar dega.

SECTION 5: BEST PRACTICES FOR ERROR REDUCTION / GALTİ KAM KRNE KA TARIKA

To minimize Sampling Error and control outlier distortion in the real world, follow these core practices:

- **Always Prefer Stratified Sampling:** Divide the population into logical strata (groups) based on known proportions before drawing random samples. This prevents extreme groups from hijacking the metrics.
- **Scale Up the Sample Size (n):** Ensure your sample size is statistically sufficient. Larger datasets naturally dilute the variance and mathematical pull of rare outliers.
- **Data Cleaning & Trimmed Metrics:** Identify extreme outliers during initial data analysis. Use tools like a Trimmed Mean (excluding the top and bottom 5% values) or shift directly to Median analysis.

Real-world analysis me Sampling Error aur Outlier ke asar ko kam se kam karne ke liye ye rules follow karein:

- **Hamesha Stratified Sampling Use Karein:** Population ko pehle unke proportions ke hisab se logical groups me baantein, phir sample chunein. Isse koi ek extreme group poore data ko kharab nahi kar payega.
- **Sample Size (n) Ko Bada Rakkhein:** Chote sample size me galti ka chance zyada hota hai. Sample size bada rakhne se ek-adh outlier ka asar poore data me dilute (khatam) ho jata hai.
- **Data Cleaning Aur Trimmed Metrics:** Processing se pehle extreme values ko scan karein. Aise cases me Trimmed Mean (top aur bottom ke 5% ko hata kar average nikalna) ya direct Median ka use karein.